

Levelling up

Everything the course assumes you already have: powers, expanding, factoring, fractions, equations, inequalities and systems.

- 1 Powers and roots
 - 2 Working with algebraic expressions
 - 3 Factoring
 - 4 Fractions
 - 5 Linear equations
 - 6 Quadratic equations
 - 7 Turning a situation into an equation
 - 8 Inequalities and percentages
 - 9 Systems of linear equations
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Powers and roots

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► **Watch:** The five rules · Exponent properties review (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:rational-exponents-radicals/x2f8bb11595b61c86:exponent-properties-review/v/multiplying-and-dividing-powers-with-integer-exponents>)

► **Watch:** Simplifying square roots (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:rational-exponents-radicals/x2f8bb11595b61c86:simplifying-square-roots/v/simplifying-square-roots-1>)

✍ **Practise:** The rules together · Properties of exponents challenge (integer exponents) (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:rational-exponents-radicals/x2f8bb11595b61c86:exponent-properties-review/e/properties-of-integer-exponents>)

✍ **Practise:** Simplify square roots (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:rational-exponents-radicals/x2f8bb11595b61c86:simplifying-square-roots/e/simplifying_radicals)

1 Same base, multiplied: add the exponents.

$$a^m \cdot a^n = a^{m+n}$$

2 Same base, divided: subtract them.

$$\frac{a^m}{a^n} = a^{m-n}$$

3 A power of a power: multiply them.

$$(a^m)^n = a^{mn}$$

4 A product inside a power: the power reaches every factor.

$$(ab)^n = a^n b^n$$

5 Zero and negative exponents are definitions, not tricks.

$$a^0 = 1 \quad (a \neq 0) \quad a^{-n} = \frac{1}{a^n}$$

6 A root is a fractional exponent, which is why the same five rules cover roots.

$$\sqrt[n]{a} = a^{1/n} \quad \text{the } n\text{-th root of } a = a^{1/n}$$

SIMPLIFY $\frac{(2x^3)^4}{4x^5}$

$$\frac{16x^{12}}{4x^5}$$

The fourth power reaches the 2 as well as the x^3 , so $2^4 = 16$ and x^{12} .

$$4x^7$$

Divide the numbers, subtract the exponents.

WATCH OUT

- $(a + b)^2$ is not $a^2 + b^2$. A power does not reach across a plus.
- a^{-n} is not negative. It is a reciprocal, so $2^{-3} = \frac{1}{8}$.
- $\sqrt{a^2 + b^2}$ does not split into $a + b$. Test it at $a = b = 3$.

Working with algebraic expressions

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► **Watch:** Expanding, from the start · Multiplying binomials intro (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:multiply-binomial/v/multiplying-simple-binomials>)

► **Watch:** Two brackets · Multiplying binomials (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:multiply-binomial/v/area-model-for-multiplying-binomials>)

≡ **Read:** Worked through · Warmup: Multiplying binomials (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:multiply-binomial/a/warmup-multiplying-binomials>)

✍ **Practise:** Expanding · Multiply binomials (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:multiply-binomial/e/multiply-binomials-coefficient>)

✍ **Practise:** The difference of two squares · Multiply difference of squares (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:special-product-binomials/e/multiplying_expressions_1)

TERM

What the + and – signs separate. In $3x^2 - 5x + 2$ there are three terms.

FACTOR

What multiplies the whole of something. Factors cancel; terms never do.

LIKE TERMS

Same letter, same power. $3x$ and $5x$ are alike; $3x$ and $3x^2$ are not.

1 Distribute across every term in the bracket.

$$a(b + c) = ab + ac$$

2 Two brackets: every term of the first against every term of the second.

$$(a + b)(c + d) = ac + ad + bc + bd$$

3 Two patterns are worth knowing by sight.

$$(a + b)(a - b) = a^2 - b^2 \quad (a \pm b)^2 = a^2 \pm 2ab + b^2$$

- 4 A minus in front of a bracket reaches every term inside it.

$$-(x - 4) = -x + 4$$

- 5 Collect like terms last. Nothing else combines.

EXPAND $(2x - 5)(3x + 4)$

$$6x^2 + 8x - 15x - 20$$

Each term of the first bracket against each term of the second.

$$6x^2 - 7x - 20$$

Collect the two x terms.

WATCH OUT

- $2(x + 3)$ is $2x + 6$, not $2x + 3$. The 2 reaches both terms.
- $(q + 3)^2$ is $q^2 + 6q + 9$. The middle term is the one people drop.
- $3x$ and $3x^2$ do not combine. Different power, different term.

Factoring

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- ▶ **Watch:** What factoring is · Introduction to factoring (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:intro-factoring/v/factors-and-divisibility-in-algebra>)
- ▶ **Watch:** Factoring a quadratic · Factoring quadratics intro (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:factor-quadratics-intro/v/factoring-simple-quadratic-expression>)
- ▶ **Watch:** The difference of two squares · Factoring quadratics with difference of squares (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:factor-difference-squares/v/difference-of-squares-intro>)
- ≡ **Read:** The method, written out · Factoring simple quadratics review (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:factor-quadratics-intro/a/factoring-simple-quadratics-review>)
- ✍ **Practise:** Common factor first · GCF factoring introduction (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:intro-factoring/e/gcf-factoring-introduction>)
- ✍ **Practise:** Factoring a quadratic · Factoring quadratics intro (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:factor-quadratics-intro/e/factoring_polynomials_1)

- 1 Take out the common factor first, every time.

$$6x^2 + 9x = 3x(2x + 3)$$

- 2 For $x^2 + bx + c$, find two numbers that multiply to c and add to b .

$$x^2 - 13x + 36 = (x - 4)(x - 9)$$

- 3 Difference of two squares.

$$a^2 - b^2 = (a - b)(a + b)$$

- 4 Perfect square.

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

- 5 **When the leading coefficient is not 1, split the middle term and group.** Find two numbers multiplying to ac and adding to b .
- 6 **Check by expanding.** It costs ten seconds and catches every sign error.

FACTOR $3x^2 + 11x - 4$

$$ac = 3 \times (-4) = -12$$

Two numbers multiplying to -12 and adding to 11 : 12 and -1 .

$$3x^2 + 12x - x - 4$$

Split the middle term using those two numbers.

$$3x(x + 4) - 1(x + 4)$$

Group in pairs and take each common factor out.

$$(3x - 1)(x + 4)$$

The bracket $x + 4$ is now itself the common factor.

WATCH OUT

- $x^2 + 9$ does not factor over the real numbers. A sum of two squares never does.
- Missing the common factor first makes every later step harder and sometimes impossible.
- $4x^2 - 16$ is $4(x - 2)(x + 2)$. Completely means completely.

Fractions

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► **Watch:** Dividing fractions by fractions (<https://www.khanacademy.org/math/arithmetic/x18ca194a:divide-fractions/x18ca194a:dividing-fractions-by-fractions/v/conceptual-understanding-of-dividing-fractions-by-fractions>)

✍ **Practise:** Adding and subtracting · Add and subtract fractions (<https://www.khanacademy.org/math/arithmetic/x18ca194a:add-and-subtract-fractions-different-denominators/x18ca194a:adding-and-subtracting-fractions-with-unlike-denominators/e/adding-and-subtracting-fractions-with-unlike-denominators>)

✍ **Practise:** Dividing fractions (https://www.khanacademy.org/math/arithmetic/x18ca194a:divide-fractions/x18ca194a:dividing-fractions-by-fractions/e/dividing_fractions_1.5)

1 **Multiply straight across. No preparation needed.**

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

2 **Divide by turning the second one over.**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

3 **Only adding and subtracting need a common denominator.** You can add halves to halves, not halves to thirds.

4 **Cancel factors, never terms.** A factor multiplies the whole numerator or the whole denominator.

5 **Keep the minus sign in front.**

$$\frac{-a}{b} = \frac{a}{-b} = -\frac{a}{b}$$

SIMPLIFY $\frac{x^2 - 25}{x + 5}$

$$\frac{(x - 5)(x + 5)}{x + 5}$$

Factor first. Only now is $x + 5$ a factor rather than a term.

$$x - 5$$

Valid for $x \neq -5$, since the original was undefined there.

WATCH OUT

- $\frac{x + 5}{x}$ is not 5. The x on top is a term, so it does not cancel.
- Test any cancellation with a number. At $x = 1$, $\frac{1 + 5}{1} = 6$, not 5.
- Adding numerators and denominators separately is the single most expensive habit on this list.

Linear equations

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► **Watch:** Unknown on both sides · Linear equations with variables on both sides (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-with-variables-both-sides/v/why-we-do-the-same-thing-to-both-sides-multi-step-equations>)

► **Watch:** With brackets · Linear equations with parentheses (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-parentheses/v/solving-equations-with-the-distributive-property>)

≡ **Read:** The method, written out · Multi-step equations review (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-parentheses/a/multi-step-equations-review>)

✍ **Practise:** Unknown on both sides · Equations with variables on both sides (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-variables-both-sides/e/linear_equations_3)

✍ **Practise:** Unknown in a denominator · Rational equations intro (https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:eq/x2ec2f6f830c9fb89:rational-eq/e/linear_equations_4)

- 1 **Whatever you do, do it to both sides.** That is the only rule; the rest is order.
- 2 **Clear brackets first.** $-(x - 4) = -x + 4$.
- 3 **Then clear fractions by multiplying every term by the common denominator.** Every term, including the ones on the right.
- 4 **Collect the unknown on one side and the numbers on the other.**
- 5 **Divide by the coefficient of the unknown.**
- 6 **If the unknown sits in a denominator, multiply through by it, then check.** A value that makes any original denominator zero is not a solution.

SOLVE $\frac{2x + 1}{3} - \frac{x - 4}{2} = 1$

$$2(2x + 1) - 3(x - 4) = 6$$

Multiply every term by 6, the common denominator.

$$4x + 2 - 3x + 12 = 6$$

Expand. The minus reaches both terms of the second bracket.

$$x + 14 = 6$$

Collect.

$$x = -8$$

Check: $\frac{-15}{3} - \frac{-12}{2} = -5 + 6 = 1.$

WATCH OUT

- Multiplying by the common denominator and forgetting the right-hand side.
- $-3(x - 4)$ is $-3x + 12$. Both signs change.
- With the unknown in a denominator, an answer that makes it zero must be thrown away.

Quadratic equations

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► **Watch:** Solving quadratics by factoring (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratics-solve-factoring/v/example-1-solving-a-quadratic-equation-by-factoring>)

► **Watch:** The formula · The quadratic formula (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/v/using-the-quadratic-formula>)

≡ **Read:** The method, written out · Solving quadratics by factoring review (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratics-solve-factoring/a/solving-quadratics-by-factoring-review>)

✍ **Practise:** Quadratics by factoring (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratics-solve-factoring/e/solving_quadratics_by_factoring_2)

✍ **Practise:** Using the formula · Quadratic formula (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/e/quadratic_equation)

1 Get everything to one side.

$$ax^2 + bx + c = 0, \quad a \neq 0$$

2 Read off a , b and c with their signs. The signs are where the marks go.

3 Try factoring. If it factors, it is faster. A product is zero only when one of its factors is zero.

4 Otherwise use the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

5 The discriminant answers how many roots before you find them.

$$\Delta = b^2 - 4ac \quad \Delta > 0: \text{two} \quad \Delta = 0: \text{one} \quad \Delta < 0: \text{none in } \mathbb{R}$$

SOLVE $2x^2 - 7x + 3 = 0$

$a = 2, \quad b = -7, \quad c = 3$

Read them off with their signs.

$\Delta = 49 - 24 = 25$

Positive, so there are two roots, and 25 is a square so they are tidy.

$x = \frac{7 \pm 5}{4}$

Put it in the formula.

$x = 3 \quad \text{or} \quad x = \frac{1}{2}$

WATCH OUT

- $b = -7$ gives $b^2 = 49$, positive. Squaring kills the sign.
- $(x - 2)(x + 5) = 4$ tells you nothing. Only a product equal to *zero* splits.
- A negative Δ means no real solution, which in a business question usually means the target is never reached.

Turning a situation into an equation

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► **Watch:** A situation into an equation · Linear equation word problems (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:forms-of-linear-equations/x2f8bb11595b61c86:intro-to-slope-intercept-form/v/linear-equation-word-problems>)

✍ **Practise:** Writing the equation · Writing linear equations word problems (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:forms-of-linear-equations/x2f8bb11595b61c86:writing-slope-intercept-equations/e/constructing-linear-functions-word-problems>)

✍ **Practise:** Writing and solving · Linear equations word problems (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:forms-of-linear-equations/x2f8bb11595b61c86:intro-to-slope-intercept-form/e/interpreting-features-of-linear-functions>)

✍ **Practise:** Two-step problems · Two-step equations word problems (<https://www.khanacademy.org/math/algebra-basics/alg-basics-linear-equations-and-inequalities/alg-basics-two-steps-equations-intro/e/linear-equation-world-problems-2>)

- 1 **Name the unknown, with its unit.** Write it down. "Let s = the number of soups sold."
- 2 **Write every other quantity in terms of it.** If 80 items sold and s are soups, the rest is $80 - s$.
- 3 **Find the sentence that says two things are equal. That is the equation.**
- 4 **Solve it.** Now it is only algebra.
- 5 **Answer in words, with the unit, and check it against the situation.** A negative count or a price of 0.003 means the setup was wrong, not the arithmetic.

A STALL SELLS ROLLS AT 4.50 AND SOUPS AT 7.00. IT SOLD 80 ITEMS AND TOOK 440.00. HOW MANY SOUPS?

Let s = soups, so rolls = $80 - s$

One unknown is enough; everything else follows from it.

$$7s + 4.5(80 - s) = 440$$

The total takings is the sentence that says two things are equal.

$$2.5s + 360 = 440$$

Expand and collect.

$$s = 32$$

32 soups and 48 rolls. Check: $32(7) + 48(4.5) = 224 + 216 = 440$.

WATCH OUT

- Starting to write algebra before defining the letter. Half the lost marks are here.
- Mixing units. If one price is in francs, all of them are.
- Answering with a bare number. The question asked how many soups.

Inequalities and percentages

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► **Watch: Multi-step inequalities** (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:multistep-inequalities/v/multi-step-inequalities-3>)

► **Watch: Percentages · Percent problems** (<https://www.khanacademy.org/math/pre-algebra/xb4832e56:percentages/xb4832e56:percent-problems/v/finding-percentages-example>)

≡ **Read: Double inequalities · Compound inequalities review** (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:compound-inequalities/a/compound-inequalities-review>)

✍ **Practise: Multi-step linear inequalities** (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:multistep-inequalities/e/linear_inequalities)

✍ **Practise: Percentages · Percent word problems** (https://www.khanacademy.org/math/pre-algebra/xb4832e56:percentages/xb4832e56:percent-word-problems/e/percentage_word_problems_1)

1 An inequality solves like an equation, with one exception.

2 Multiplying or dividing by a negative flips the sign.

$$-2x > 6 \quad \Rightarrow \quad x < -3$$

3 A double inequality: do the same thing to all three parts.

$$2 \leq 3x - 1 \leq 8 \quad \Rightarrow \quad 1 \leq x \leq 3$$

4 A percentage is a multiplier. That is the whole idea.

$$p\% \text{ of } A = \frac{p}{100} \cdot A$$

5 Up by $p\%$ multiplies; down by $p\%$ multiplies too.

$$\times \left(1 + \frac{p}{100}\right) \quad \times \left(1 - \frac{p}{100}\right)$$

6 Successive changes multiply. They never add.

A PRICE RISES 20%, THEN FALLS 20%. WHAT IS THE NET CHANGE?

×1.20 then ×0.80

Each change is a multiplier on what came before.

$$1.20 \times 0.80 = 0.96$$

Multiply the multipliers.

a 4% fall

Not zero. The fall is taken on the larger price.

WATCH OUT

- $-2x > 6$ gives $x < -3$. Dividing by the negative flips it.
- Up 20% then down 20% is not back where you started.
- A percentage point is not a percent. From 4% to 6% is two points and a 50% rise.

Systems of linear equations

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► **Watch:** By substitution · Solving systems of equations with substitution (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:systems-of-equations/x2f8bb11595b61c86:solving-systems-of-equations-with-substitution/v/solving-systems-with-substitution>)

✍ **Practise:** By substitution · Systems of equations with substitution (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:systems-of-equations/x2f8bb11595b61c86:solving-systems-of-equations-with-substitution/e/systems_of_equations_with_substitution)

✍ **Practise:** By elimination · Systems of equations with elimination (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:systems-of-equations/x2f8bb11595b61c86:solving-systems-elimination/e/systems_of_equations_with_elimination_0.5)

- 1 **Two unknowns need two independent equations.** Independent means neither is a multiple of the other.
- 2 **Substitution: make one equation say $x = \text{something}$, then put that into the other.** The other one. Putting it back into the equation you rearranged gives $0 = 0$.
- 3 **Elimination: scale one or both so a letter cancels when you add or subtract.**
- 4 **If the letters vanish and a false statement is left, there is no solution.** $0 = 5$. The lines are parallel.
- 5 **If the letters vanish and a true statement is left, there are infinitely many.** $0 = 0$. It was the same line twice.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

SOLVE $3x + 2y = 19$ AND $x - y = 3$

$$x = y + 3$$

Rearrange the simpler of the two.

$$3(y + 3) + 2y = 19$$

Substitute into the *other* equation.

$$5y = 10, \quad y = 2$$

Expand and collect.

$$x = 5, \quad y = 2$$

Check both: $15 + 4 = 19$ and $5 - 2 = 3$.

WATCH OUT

- Substituting back into the equation you just rearranged. It always gives $0 = 0$.
- Subtracting equations and forgetting that the signs on the right change too.
- Checking in one equation only. A slip survives that.



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